

Machine Learning Roadmap

You are my **Machine Learning Mentor, Data Scientist Interview Trainer, and Case Study Instructor**.

I am following a **60-day / 2-month Machine Learning roadmap** to become **job-ready for a Data Scientist role**.

My goal is to learn Machine Learning from **beginner** → **intermediate** → **advanced** → **practical Data Science** → **interview level**.

Show moreShow less

Absolutely. I'll act as your **Machine Learning Mentor + Data Scientist Interview Trainer + Case Study Instructor** throughout the 60-day program.

We'll follow a structured **Beginner** → **Intermediate** → **Advanced** → **Real-world Data Science** → **Interview-ready** progression.

How each daily lesson will work

For every day of the 60-day roadmap, I'll cover:

1. Concept & Theory

- Intuition first
- Mathematical foundations where necessary
- Important terminology
- When/why to use the technique

2. Python Coding

- From simple examples → practical implementation
- NumPy/Pandas
- Matplotlib/Seaborn where relevant
- Scikit-learn
- Model training, evaluation and interpretation

3. Data Science Case Study

- Business problem
- Dataset understanding
- EDA
- Feature engineering
- Model development
- Evaluation
- Business recommendations

4. Interview Preparation

- Conceptual questions
- Coding questions
- Scenario-based questions
- ML system/business questions
- Model-selection questions
- **Top-company-style interview questions with answers**

5. 30 MCQs

- Beginner → intermediate → advanced
- Answers
- Explanations
- Interview-oriented traps

6. Practical Coding Challenge

- A small task related to the day's topic
- Progressively harder throughout the 60 days

7. End-of-day assessment

- What you should know
- What you should be able to code
- Common mistakes
- Interview readiness checklist

60-Day progression

Phase	Days	Focus
Phase 1	1–10	ML foundations, Python/NumPy/Pandas refresh, statistics, probability, data preprocessing
Phase 2	11–20	Linear Regression, Logistic Regression, KNN, Naive Bayes
Phase 3	21–30	Decision Trees, Random Forest, Ensemble Learning, SVM
Phase 4	31–40	Unsupervised Learning, Clustering, PCA, anomaly detection, feature engineering
Phase 5	41–50	Advanced ML, XGBoost/LightGBM, hyperparameter tuning, pipelines, model interpretation
Phase 6	51–60	End-to-end projects, deployment concepts, ML system design, business cases, interview preparation

By the end, the target is not merely **"I know ML algorithms."** It is:

Business problem → Data → EDA → Feature Engineering → Model → Evaluation → Interpretation → Business Decision → Deployment/Monitoring

That is the workflow I'll train you to explain in a Data Scientist interview.

Send START DAY 1 when you're ready, and I'll begin the first lesson.

1. Explain what Machine Learning is.
2. Explain AI vs ML vs Deep Learning.
3. Explain traditional programming vs Machine Learning.
4. Explain why Machine Learning is useful.
5. Understand the major types of Machine Learning.
6. Understand supervised learning.
7. Understand unsupervised learning.
8. Understand semi-supervised learning.
9. Understand reinforcement learning.
10. Understand regression vs classification.
11. Understand features and target variables.
12. Understand observations, samples, labels, and predictions.
13. Understand training data and testing data.
14. Understand the basic ML workflow.
15. Understand the difference between training, validation, and testing.
16. Understand overfitting and underfitting at a basic level.
17. Understand what a Machine Learning model actually learns.
18. Understand the difference between a parameter and a hyperparameter.
19. Understand how a Data Scientist approaches an ML problem.
20. Build my first Machine Learning model in Python.

Show moreShow less

Day 1 — Machine Learning Fundamentals

Today is about building the **mental model of Machine Learning** before learning algorithms.

By the end of Day 1, you should be able to explain ML to an interviewer, identify the type of ML problem, understand datasets and model training, and build your first model in Python.

1. What is Machine Learning?

Simple definition

Machine Learning is a method of teaching computers to learn patterns from data and use those patterns to make predictions or decisions without explicitly programming every rule.

Traditional programming:

Rules + Data → Output

Machine Learning:

Data + Correct Outputs → Learning Algorithm → Model

Then:

New Data + Model → Prediction

Example: House Price Prediction

Suppose we have:

Area (sq ft)	Bedrooms	Location	Price
800	2	Mumbai	₹80L
1200	3	Mumbai	₹120L
1500	3	Pune	₹100L
2000	4	Mumbai	₹200L

We give this historical data to an ML algorithm.

The algorithm learns relationships such as:

- larger houses tend to cost more
- more bedrooms may increase price
- location affects price

Then we give:

Area = 1400
Bedrooms = 3
Location = Mumbai

The model might predict:

₹145 Lakhs

The important point:

We did not manually write a rule saying "every additional 100 sq ft adds ₹X."

The model learned the relationship from data.

2. AI vs ML vs Deep Learning

Think of these as nested concepts:

```

Artificial Intelligence
├── Machine Learning
│   └── Deep Learning
    
```

Artificial Intelligence

AI is the broadest field.

It focuses on creating systems capable of performing tasks that normally require human-like intelligence.

Examples:

- reasoning
 - planning
 - computer vision
 - speech recognition
 - recommendation
 - language understanding
 - decision-making
-

Machine Learning

ML is a **subset of AI**.

Instead of explicitly programming every rule, we allow algorithms to learn patterns from data.

Examples:

- spam detection
 - credit-risk prediction
 - customer churn prediction
 - fraud detection
 - stock-price forecasting
 - recommendation systems
-

Deep Learning

Deep Learning is a **subset of Machine Learning** based primarily on neural networks with multiple layers.

Examples:

- image recognition
- speech recognition
- LLMs
- autonomous driving perception
- medical image analysis

Interview answer

AI is the broad field of building intelligent systems. Machine Learning is a subset of AI where systems learn patterns from data. Deep Learning is a subset of ML that primarily uses multi-layer neural networks to learn complex representations from large datasets.

3. Traditional Programming vs Machine Learning

Traditional programming

Imagine a bank wants to determine whether a transaction is fraudulent.

A traditional approach might use manually written rules:

```
IF transaction_amount > ₹100000
AND transaction_location != customer's usual location
THEN fraud
```

The programmer defines the rules.

```
INPUT DATA
+
PROGRAMMED RULES
↓
OUTPUT
```

Machine Learning

Instead, give the algorithm historical transactions:

Amount	Location Change	Time	Previous Fraud	Fraud
500	No	10 AM	No	No
150000	Yes	2 AM	No	Yes
2000	No	6 PM	No	No
90000	Yes	3 AM	Yes	Yes

The algorithm learns patterns.

```
Historical Data
+
Correct Answers
↓
ML Algorithm
↓
Trained Model
↓
New Transaction
↓
Fraud / Not Fraud
```

4. Why is Machine Learning useful?

ML becomes particularly useful when:

1. Rules are difficult to write

Example:

Is this image a cat or dog?

Writing thousands of visual rules manually is impractical.

2. Patterns are complex

Example:

Credit risk depends on:

- income
- age
- existing loans
- repayment history
- utilization
- employment
- transaction behavior

The interactions can be difficult to express manually.

3. Large amounts of data exist

Banks, e-commerce companies, social media platforms and financial institutions generate huge datasets.

ML can discover patterns at scale.

4. Predictions are required

Examples:

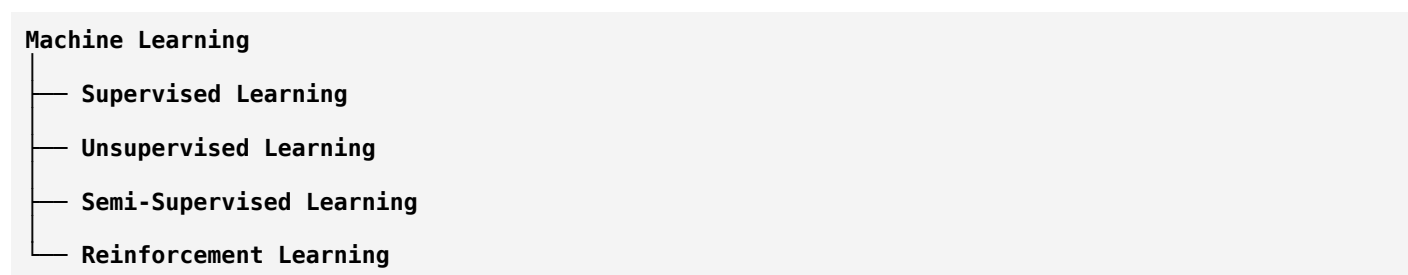
```
Will customer churn?  
Will loan default?  
What will the house price be?  
Is this transaction fraudulent?  
Which product should we recommend?
```

5. Systems need to improve from data

A recommendation system can continually learn from new user behavior.

5. Major Types of Machine Learning

The four major categories you'll encounter are:



6. Supervised Learning

In supervised learning, we have:

Input data + known target/output

The model learns a mapping:

$$X \rightarrow Y$$

Where:

- X = features
- Y = target

Example

Predict whether a customer will churn.

Age	Salary	Tenure	Churn
25	40000	1	Yes
35	80000	5	No
42	90000	8	No

The model learns from examples where the correct answer is known.

Common supervised algorithms

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- SVM
- XGBoost
- LightGBM
- Neural Networks

7. Unsupervised Learning

Here we **don't have a target variable**.

The model tries to discover hidden patterns or structures.

X → Algorithm → Hidden Structure

Example: Customer Segmentation

Suppose we have:

Customer	Income	Spending
A	30K	5K
B	35K	6K
C	150K	50K
D	160K	55K

There is no column saying:

Customer Group

The algorithm may discover:

Cluster 1 → Low income / low spending
Cluster 2 → High income / high spending

Common algorithms

- K-Means
- Hierarchical Clustering
- DBSCAN
- PCA
- Gaussian Mixture Models

8. Semi-Supervised Learning

Here we have:

Small amount of labeled data + large amount of unlabeled data

Example:

10,000 images
 └─ **1,000 labeled**
 └─ **9,000 unlabeled**

Labeling every image manually could be expensive.

Semi-supervised methods use the labeled and unlabeled data together.

Useful when:

- labeling is expensive
- unlabeled data is abundant

9. Reinforcement Learning

Reinforcement Learning is based on:

Agent → **Environment** → **Action** → **Reward/Penalty**

Example: game-playing AI.



Suppose an AI plays chess.

Good move:

+10 reward

Bad move:

-10 reward

The agent learns a strategy that maximizes cumulative reward.

Applications:

- robotics
- games
- autonomous systems
- recommendation optimization
- resource allocation

10. Regression vs Classification

This is extremely important for interviews.

Regression

Regression predicts a **continuous numerical value**.

Examples:

```
House price → ₹1.25 crore
Salary → ₹12.5 lakh
Temperature → 32.5°C
Sales → ₹2.4 million
```

Example:

```
Input:
Area = 1200 sq ft

Output:
₹1.1 crore
```

Common algorithms:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting
- XGBoost

Classification

Classification predicts a **category/class**.

Examples:

```
Spam / Not Spam
Fraud / Not Fraud
Churn / Not Churn
Disease / No Disease
Default / No Default
```

Example:

```
Customer data
↓
Model
↓
Churn = Yes
```

Binary classification

Two classes:

```
Yes / No
0 / 1
```

Fraud / Not Fraud

Multiclass classification

More than two classes:

Low
Medium
High

or:

Cat
Dog
Horse
Bird

11. Features and Target Variables

Suppose:

Age	Income	Credit Score	Loan Amount	Default
25	40000	650	500000	No
42	90000	780	300000	No
31	35000	580	700000	Yes

Features

The input variables:

Age
Income
Credit Score
Loan Amount

Usually represented as:

$$X$$

Target

The value we want to predict:

Default

Represented as:

$$Y$$

So:

$$X = [Age, Income, CreditScore, LoanAmount]$$

12. Observations, Samples, Labels and Predictions

These terms are often confusing.

Observation / Sample

One row of data.

```
25 | 40000 | 650 | 500000 | No
```

That's one observation.

Dataset

Collection of observations.

```
10,000 rows
```

means we have 10,000 observations.

Label

The known target value in supervised learning.

Example:

```
Default = Yes
```

is the label.

Prediction

The model's estimated output for new data.

Actual:

```
Default = Yes
```

Model prediction:

```
Default = No
```

The prediction was wrong.

13. Training Data and Testing Data

We shouldn't train and evaluate our model on exactly the same data.

Suppose:

10,000 records

We might divide them:

Training → 8,000
Testing → 2,000

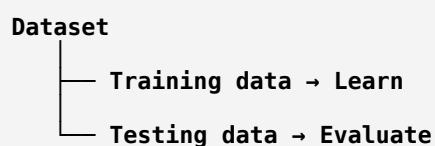
Training set

Used to learn patterns.

Test set

Used to evaluate how well the trained model performs on **unseen data**.

Conceptually:



14. Basic ML Workflow

This workflow is extremely important for a Data Scientist.

1. Business Problem
- ↓
2. Collect Data
- ↓
3. Understand Data
- ↓
4. Clean Data
- ↓
5. EDA
- ↓
6. Feature Engineering
- ↓
7. Train / Validation / Test Split
- ↓
8. Model Selection
- ↓
9. Train Model
- ↓
10. Evaluate Model
- ↓
11. Tune Model
- ↓
12. Interpret Results
- ↓
13. Deploy
- ↓
14. Monitor

Notice that **model training is only one part of the ML lifecycle**.

15. Training vs Validation vs Testing

This is a very common interview question.

Suppose:

100,000 records

Possible split:

**70% Training
15% Validation
15% Testing**

Training set

Used to learn model parameters.

Training → Model learns

Validation set

Used for decisions such as:

- model selection
- hyperparameter tuning
- threshold selection

Validation → Choose/tune model

Test set

Used for final unbiased evaluation.

Test → Final performance estimate

Important principle

Don't repeatedly tune your model based on the test set.

Otherwise, the test set gradually influences your decisions and is no longer a clean estimate of generalization.

16. Overfitting and Underfitting

Imagine you're studying for an exam.

Underfitting

You studied too little.

You don't understand the concepts.

Training Performance → Poor
Testing Performance → Poor

The model is too simple or hasn't learned enough.

Good fit

Training Performance → Good
Testing Performance → Good

The model captures useful general patterns.

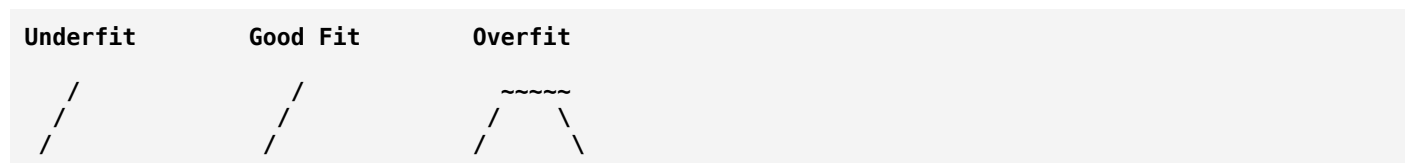
Overfitting

You memorized the training questions instead of understanding the concepts.

Training Performance → Excellent
Testing Performance → Poor

The model has learned noise/details specific to the training data.

Visual intuition:



We'll study methods such as:

- regularization
- cross-validation
- pruning
- dropout
- early stopping
- feature selection

later in the roadmap.

17. What does an ML model actually learn?

This is one of the most important concepts.

Suppose we have Linear Regression:

$$y = w_1x_1 + w_2x_2 + b$$

The model needs to learn:

```
w1
w2
b
```

These are **parameters**.

Example:

```
House Price =
₹80,000 × Area
+ ₹10,00,000 × Bedrooms
+ Location Effect
+ ...
```

The model doesn't "understand" the house like a human.

It learns **numerical relationships/patterns in the training data according to the model's structure and objective**.

During training:

```
Input
↓
Model
↓
Prediction
↓
Compare with actual
↓
Calculate error/loss
↓
Update parameters
↓
Repeat
```

Eventually, the algorithm finds parameter values that minimize the chosen loss/objective.

18. Parameter vs Hyperparameter

Another important interview question.

Parameter

A value **learned from the training data**.

Example in Linear Regression:

```
w1
w2
b
```

The model learns these.

Hyperparameter

A value **chosen before/during training by the practitioner or tuning procedure**, rather than directly learned as a model parameter.

Examples:

```
Learning rate
Number of trees
Tree depth
K in KNN
Regularization strength
```

For example:

```
Python
RandomForestClassifier(
    n_estimators=100,
    max_depth=10
)
```

Here:

```
n_estimators → hyperparameter
max_depth → hyperparameter
```

The individual tree split values and other learned quantities are parameters/model state learned from the training data.

Easy memory trick

Parameters = model learns them.

Hyperparameters = we configure/tune them.

19. How does a Data Scientist approach an ML problem?

A Data Scientist doesn't start by saying:

"Let's use Random Forest."

Instead:

Step 1 — Understand the business problem

Example:

Bank wants to reduce loan defaults.

Step 2 — Define the ML problem

Can we predict:

Default = Yes/No

This becomes a **binary classification** problem.

Step 3 — Define success

Maybe:

Precision $\geq$ 80%
Recall $\geq$ 75%

depending on the business cost of false positives vs false negatives.

Step 4 — Collect data

Potential variables:

Income
Credit score
Loan amount
Employment
Existing debt
Payment history
Account behavior

Step 5 — Explore and clean

Check:

- missing values
- outliers
- duplicates
- incorrect data
- class imbalance
- data leakage

Step 6 — Engineer features

Create meaningful variables.

Example:

$$Debt\ Ratio = \frac{Total\ Debt}{Income}$$

Step 7 — Split data

Train
Validation
Test

Step 8 — Build baseline

Start simple.

For example:

```
Logistic Regression
```

Step 9 — Try stronger models

```
Random Forest
XGBoost
LightGBM
```

Step 10 — Evaluate

Use metrics appropriate to the problem.

Step 11 — Interpret

Ask:

Why is the model predicting default?

Step 12 — Business decision

Finally:

Should the bank approve, reject, or manually review this application?

That's Data Science.

20. Your First Machine Learning Model

Let's build a simple **House Price Prediction** model.

We'll use a small synthetic dataset so you understand every step.

Step 1 — Import libraries

```
Python
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

Step 2 — Create dataset

```
Python
```

```
data = {
    "area_sqft": [600, 800, 1000, 1200, 1400, 1600, 1800, 2000],
    "bedrooms": [1, 2, 2, 3, 3, 3, 4, 4],
    "price_lakh": [45, 60, 72, 90, 105, 120, 140, 155]
}

df = pd.DataFrame(data)

print(df)
```

We have:

```
Features:
area_sqft
bedrooms

Target:
price_lakh
```

Step 3 — Separate X and y

```
Python
X = df[["area_sqft", "bedrooms"]]
y = df["price_lakh"]
```

Remember:

```
X → inputs/features
y → target
```

Step 4 — Train/Test Split

```
Python
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)
```

We are keeping:

```
75% → Training
25% → Testing
```

`random_state=42` makes the split reproducible.

Step 5 — Create the model

```
Python
model = LinearRegression()
```

At this point, the model has **not learned from our dataset yet**.

Step 6 — Train

```
Python
model.fit(X_train, y_train)
```

This is where learning occurs.

Conceptually:

```
Training data
  ↓
Linear Regression
  ↓
Learn coefficients
```

Step 7 — Make predictions

```
Python
y_pred = model.predict(X_test)
print(y_pred)
```

Step 8 — Evaluate

Mean Absolute Error

```
Python
mae = mean_absolute_error(y_test, y_pred)
print("MAE:", mae)
```

If:

```
MAE = 5
```

the model's predictions are off by about **5 lakh on average**, in absolute terms.

R^2

```
Python
r2 = r2_score(y_test, y_pred)
print("R²:", r2)
```

R^2 gives an indication of how much variation in the target is explained by the model under the assumptions of this metric.

Step 9 — Predict a new house

Suppose we want to predict:

```
Area = 1500 sq ft
Bedrooms = 3
```

Python

```
new_house = pd.DataFrame({
    "area_sqft": [1500],
    "bedrooms": [3]
})

prediction = model.predict(new_house)

print("Predicted price:", prediction[0], "lakh")
```

You've just built your **first Machine Learning model**.

The Complete Code

Python

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

# 1. Dataset
data = {
    "area_sqft": [600, 800, 1000, 1200, 1400, 1600, 1800, 2000],
    "bedrooms": [1, 2, 2, 3, 3, 3, 4, 4],
    "price_lakh": [45, 60, 72, 90, 105, 120, 140, 155]
}

df = pd.DataFrame(data)

# 2. Features and target
X = df[["area_sqft", "bedrooms"]]
y = df["price_lakh"]

# 3. Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)

# 4. Create model
model = LinearRegression()

# 5. Train
model.fit(X_train, y_train)

# 6. Predict
y_pred = model.predict(X_test)
```

```
# 7. Evaluate
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("R²:", r2)

# 8. New prediction
new_house = pd.DataFrame({
    "area_sqft": [1500],
    "bedrooms": [3]
})

prediction = model.predict(new_house)
print("Predicted price:", prediction[0], "lakh")
```

Day 1 Case Study — Bank Loan Default

Let's connect today's concepts to a real Data Scientist problem.

Business problem

A bank wants to identify customers likely to default on a loan.

Dataset

Suppose:

Age	Income	Credit Score	Loan Amount	Debt Ratio	Default
25	35K	580	700K	0.65	Yes
40	90K	760	300K	0.25	No
31	45K	620	500K	0.55	Yes
50	120K	800	400K	0.20	No

Identify the components

Observations: rows

Features:

```
Age
Income
Credit Score
Loan Amount
Debt Ratio
```

Target:

```
Default
```

Problem type:

```
Supervised Learning
↓
```


Classification
↓
Binary Classification

Because:

Default = Yes / No

Possible business objective

Predict high-risk applicants early enough for the bank to:

- reject the application
- reduce the loan amount
- increase scrutiny
- request additional documentation
- route the application for manual review

Notice how ML connects to a **business decision**, not just a model score.

Interview Questions — Day 1

Q1. What is Machine Learning?

Answer:

Machine Learning is a subset of AI where algorithms learn patterns from data to make predictions or decisions without requiring explicit rules for every possible situation.

Q2. What is the difference between AI and ML?

Answer:

AI is the broader field of creating systems capable of intelligent behavior, while ML is a subset of AI where systems learn patterns from data.

Q3. What is the difference between ML and Deep Learning?

Answer:

Deep Learning is a subset of Machine Learning that primarily uses multi-layer neural networks to learn complex representations from data.

Q4. What is supervised learning?

Answer:

Supervised learning learns a mapping between input features and a known target using labeled training data.

Q5. What is unsupervised learning?

Answer:

Unsupervised learning works primarily with data without target labels and attempts to discover underlying patterns, structures, or groups.

Q6. What is the difference between regression and classification?

Answer:

Regression predicts continuous numerical values, while classification predicts discrete classes or categories.

Q7. Give an example of regression.

Answer:

Predicting house prices.

Q8. Give an example of classification.

Answer:

Predicting whether a customer will default on a loan.

Q9. What is a feature?

Answer:

A feature is an input variable used by a machine learning model to make a prediction.

Q10. What is a target variable?

Answer:

The target is the output variable that a supervised learning model is trained to predict.

Q11. Why do we split data into training and testing sets?

Answer:

To evaluate whether the model generalizes to unseen data rather than simply measuring how well it memorized the training data.

Q12. What is overfitting?

Answer:

Overfitting occurs when a model learns the training data too closely, including noise or dataset-specific patterns, resulting in poor performance on unseen data.

Q13. What is underfitting?

Answer:

Underfitting occurs when a model is too simple or insufficiently trained to capture the underlying patterns, causing poor performance on both training and unseen data.

Q14. What is a parameter?

Answer:

A parameter is a model value learned from training data, such as coefficients in linear regression.

Q15. What is a hyperparameter?

Answer:

A hyperparameter is a configuration value selected before or during model training, usually through experimentation or tuning, such as tree depth or learning rate.



Day 1 — 30 MCQs

1. Machine Learning is primarily a subset of:

- A. Database Management
- B. Artificial Intelligence
- C. Operating Systems
- D. Networking

Answer: B

2. Deep Learning is a subset of:

- A. AI only
- B. ML
- C. SQL
- D. Statistics only

Answer: B

3. Predicting house price is generally:

- A. Classification
- B. Regression
- C. Clustering
- D. Reinforcement learning

Answer: B

4. Predicting spam/not-spam is:

- A. Regression
- B. Classification
- C. Clustering
- D. PCA

Answer: B

5. In supervised learning, we have:

- A. Only inputs
- B. Inputs and known targets
- C. No data
- D. Only predictions

Answer: B

6. K-Means is primarily:

- A. Supervised
- B. Unsupervised
- C. Reinforcement
- D. Regression

Answer: B

7. The input variables are called:

- A. Targets
- B. Features
- C. Labels
- D. Predictions

Answer: B

8. The variable being predicted is called:

- A. Feature
- B. Target
- C. Observation
- D. Hyperparameter

Answer: B

9. One row of a dataset usually represents:

- A. Hyperparameter
- B. Observation/sample
- C. Model
- D. Algorithm

Answer: B

10. A label is:

- A. Known target associated with an example
- B. Always a feature
- C. Always continuous
- D. A hyperparameter

Answer: A

11. Training data is primarily used to:

- A. Deploy the model
- B. Learn model parameters

- C. Delete data
- D. Generate reports only

Answer: B

12. Test data should ideally be:

- A. Used repeatedly for tuning
- B. Unseen during model development
- C. Always larger than training data
- D. Identical to training data

Answer: B

13. Validation data is commonly used for:

- A. Hyperparameter/model selection
- B. Database storage
- C. Data deletion
- D. Label creation only

Answer: A

14. A model that performs extremely well on training but poorly on test data is likely:

- A. Underfitting
- B. Overfitting
- C. Normalizing
- D. Clustering

Answer: B

15. A model performing poorly on both training and test data may be:

- A. Overfitting
- B. Underfitting
- C. Perfectly fitted
- D. Data encrypted

Answer: B

16. Which is a parameter?

- A. Learning rate
- B. Maximum tree depth
- C. Linear regression coefficient
- D. Number of trees

Answer: C

17. Which is a hyperparameter?

- A. Linear regression coefficient
- B. Learned neural network weight
- C. Tree depth
- D. Predicted value

Answer: C

18. Reinforcement Learning uses:

- A. Rewards and penalties
- B. Only labeled datasets
- C. Only clusters
- D. SQL queries

Answer: A

19. Binary classification has:

- A. One class
- B. Two classes
- C. Exactly three classes
- D. Continuous outputs

Answer: B

20. Multiclass classification has:

- A. Multiple discrete classes
- B. Only two classes
- C. Continuous output only
- D. No target

Answer: A

21. Which is an example of unsupervised learning?

- A. Spam detection
- B. House-price prediction
- C. Customer clustering
- D. Loan default prediction

Answer: C

22. Which algorithm is commonly used for regression?

- A. Linear Regression
- B. K-Means
- C. DBSCAN
- D. PCA

Answer: A

23. Which algorithm is commonly used for clustering?

- A. Linear Regression
- B. K-Means
- C. Logistic Regression
- D. Linear SVM only

Answer: B

24. In an ML workflow, EDA means:

- A. Exploratory Data Analysis
- B. External Data Algorithm
- C. Estimated Data Accuracy
- D. Embedded Data Application

Answer: A

25. Which should generally happen before model training?

- A. Data preparation
- B. Model deployment
- C. Production monitoring
- D. Final reporting only

Answer: A

26. The output generated by a trained model for new data is called:

- A. Parameter
- B. Prediction
- C. Feature
- D. Training label

Answer: B

27. Semi-supervised learning uses:

- A. Only labeled data
- B. Only unlabeled data

- C. Both labeled and unlabeled data
- D. No data

Answer: C

28. Which problem is most naturally a classification problem?

- A. Predict temperature
- B. Predict house price
- C. Predict customer churn
- D. Predict revenue

Answer: C

29. Which sequence is most appropriate?

- A. Deploy → collect data → define problem
- B. Business problem → data → preprocessing → modeling → evaluation
- C. Model → business problem → data
- D. Testing → data collection → problem definition

Answer: B

30. The ultimate goal of a Data Scientist is usually:

- A. Use the most complicated algorithm
- B. Maximize code length
- C. Solve a business/problem objective using data
- D. Always achieve 100% accuracy

Answer: C



Day 1 Coding Challenge

Build a **student exam-score prediction model**.

Dataset:

```
Python
data = {
    "hours_studied": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "attendance": [60, 65, 70, 72, 75, 80, 82, 85, 90, 95],
    "exam_score": [35, 40, 48, 52, 58, 65, 68, 75, 82, 90]
}
```

Your tasks:

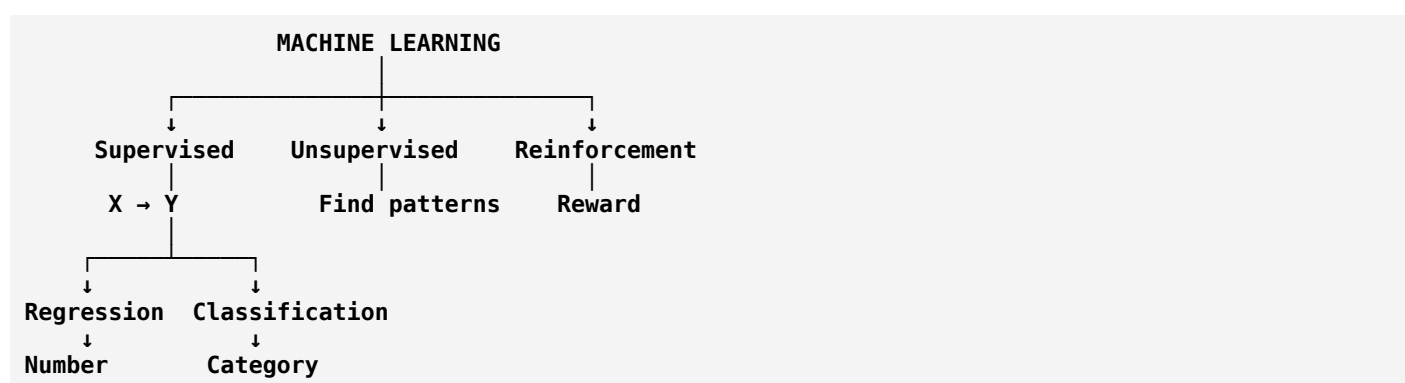
1. Create a DataFrame.
2. Identify x .
3. Identify y .

4. Split into training/testing data.
5. Create a **LinearRegression** model.
6. Train it.
7. Predict test scores.
8. Calculate MAE.
9. Calculate R^2 .
10. Predict the score for:

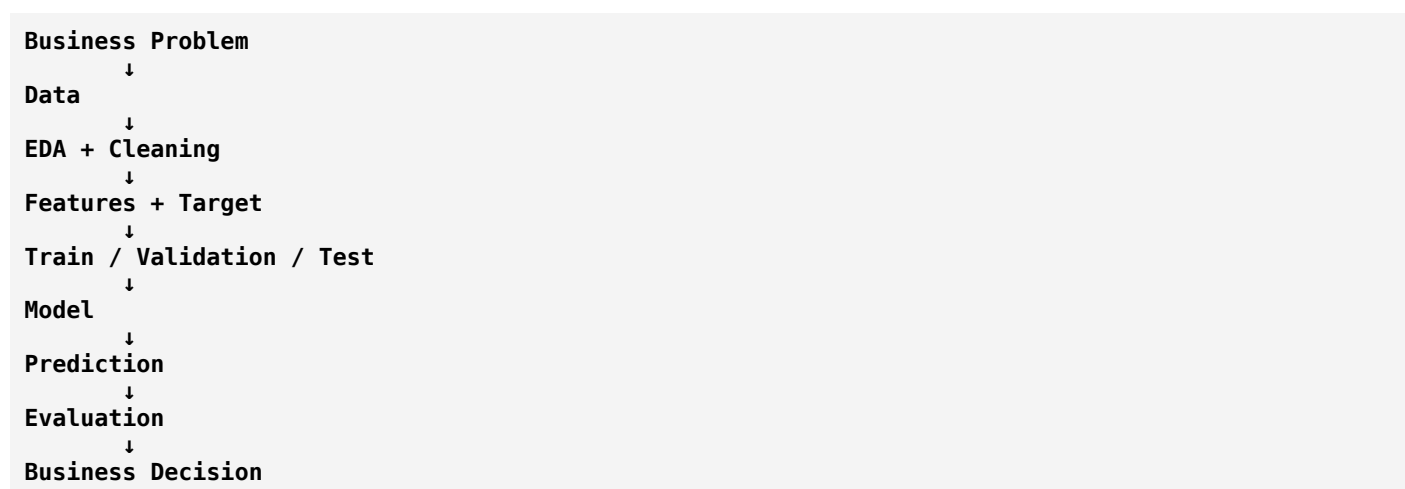
Hours studied = 7
Attendance = 88%

Your Day 1 mental model

Memorize this:



And the Data Scientist workflow:



Day 1 success criterion: You should now be able to look at a business problem such as *customer churn*, *loan default*, *fraud detection*, *house-price prediction*, or *customer segmentation* and correctly identify **the ML type, features, target, training data, and evaluation approach** before writing any model code.

30 mcq

Absolutely. Here is a **fresh set of 30 Day 1 Machine Learning MCQs**. Try answering them **without looking at the answers first**.

Day 1 — Machine Learning Fundamentals: 30 MCQs

1. Which statement best describes Machine Learning?

- A. Writing explicit rules for every possible situation
 - B. Allowing computers to learn patterns from data
 - C. Storing large amounts of data
 - D. Designing computer hardware
-

2. Which relationship is correct?

- A. $ML \subset AI \subset \text{Deep Learning}$
 - B. $AI \subset ML \subset \text{Deep Learning}$
 - C. $\text{Deep Learning} \subset ML \subset AI$
 - D. $AI = ML = \text{Deep Learning}$
-

3. Netflix recommending movies based on user behavior is an example of:

- A. Machine Learning
 - B. Operating Systems
 - C. Traditional programming only
 - D. Database normalization
-

4. In traditional programming, the programmer generally provides:

- A. Data and rules
 - B. Only data
 - C. Only outputs
 - D. Predictions and labels
-

5. In supervised learning, the model learns from:

- A. Unlabeled data only
 - B. Labeled data
 - C. Random numbers
 - D. Rewards only
-

6. Which is an example of supervised learning?

- A. Customer segmentation without predefined groups
- B. Predicting whether a customer will churn

- C. Finding hidden groups of customers
 - D. Compressing data using PCA
-

7. Which is an example of unsupervised learning?

- A. Predicting loan default
 - B. Predicting salary
 - C. Customer clustering
 - D. Predicting house price
-

8. Which type of learning uses rewards and penalties?

- A. Supervised learning
 - B. Unsupervised learning
 - C. Reinforcement learning
 - D. Semi-supervised learning
-

9. Semi-supervised learning typically involves:

- A. Only labeled data
 - B. Only unlabeled data
 - C. A combination of labeled and unlabeled data
 - D. No training data
-

10. Predicting the selling price of a house is a:

- A. Classification problem
 - B. Regression problem
 - C. Clustering problem
 - D. Reinforcement learning problem
-

11. Predicting whether an email is spam or not spam is:

- A. Regression
 - B. Binary classification
 - C. Clustering
 - D. Dimensionality reduction
-

12. Predicting whether a customer belongs to Basic, Premium, or Enterprise is:

- A. Binary classification
 - B. Multiclass classification
 - C. Regression
 - D. Clustering
-

13. In an ML dataset, the variables used as inputs are called:

- A. Labels
 - B. Features
 - C. Predictions
 - D. Errors
-

14. Consider:

Age	Income	Credit Score	Default
25	40000	650	Yes

What is the target variable?

- A. Age
 - B. Income
 - C. Credit Score
 - D. Default
-

15. One row in a typical tabular dataset is called:

- A. Hyperparameter
 - B. Observation/sample
 - C. Algorithm
 - D. Feature set
-

16. A known target value associated with a training example is called:

- A. Label
 - B. Feature
 - C. Hyperparameter
 - D. Prediction
-

17. What is the primary purpose of training data?

- A. Final production monitoring
 - B. Learning model parameters
 - C. Creating dashboards
 - D. Testing deployment infrastructure
-

18. Why do we use a test dataset?

- A. To train the model
- B. To increase the number of features
- C. To estimate performance on unseen data
- D. To remove missing values

19. What is the primary purpose of a validation dataset?

- A. Hyperparameter/model selection and tuning
 - B. Permanent model storage
 - C. Data collection
 - D. Replacing the training set
-

20. Which sequence is most appropriate?

- A. Test → Train → Validation
 - B. Train → Validation → Test
 - C. Validation → Test → Train
 - D. Test → Validation → Train
-

21. A model achieves 99% training accuracy but 65% test accuracy. What is the most likely issue?

- A. Underfitting
 - B. Overfitting
 - C. Clustering
 - D. Normalization
-

22. A model performs poorly on both training and test data. This is commonly associated with:

- A. Overfitting
 - B. Underfitting
 - C. Data leakage
 - D. Perfect generalization
-

23. Which statement about model parameters is correct?

- A. They are always manually selected
 - B. They are learned from training data
 - C. They are always equal to hyperparameters
 - D. They are never numerical
-

24. Which is a hyperparameter?

- A. Linear regression coefficient learned during training
 - B. Neural-network weight learned during training
 - C. Maximum depth of a decision tree
 - D. Predicted house price
-

25. Which of the following is part of a typical ML workflow?

- A. Business problem → Data → EDA → Modeling → Evaluation
 - B. Modeling → Business problem → Data
 - C. Deployment → Data collection → Problem definition
 - D. Testing → Coding → Data collection
-

26. A Data Scientist wants to predict whether a bank customer will default on a loan.

The target is:

`Default = Yes / No`

What type of problem is this?

- A. Regression
 - B. Binary classification
 - C. Clustering
 - D. Dimensionality reduction
-

27. A Data Scientist wants to group customers based on spending behavior without predefined labels.

Which approach is most appropriate?

- A. Supervised classification
 - B. Regression
 - C. Unsupervised clustering
 - D. Reinforcement learning
-

28. What does an ML model primarily learn during training?

- A. The exact rows of the dataset only
 - B. Useful patterns/relationships represented by its learned parameters
 - C. Business requirements
 - D. The programming language
-

29. Which approach is generally best when solving a real-world ML problem?

- A. Immediately select the most complex algorithm
 - B. Start with the business problem and define the objective
 - C. Always use Deep Learning
 - D. Optimize accuracy before understanding the data
-

30. Which statement is TRUE?

- A. A highly complex model is always better
- B. Test data should be repeatedly used to tune the model
- C. High training performance alone proves good generalization
- D. A model should be evaluated on data representative of unseen cases

✓ Answer Key

Q	Answer	Q	Answer	Q	Answer
1	B	11	B	21	B
2	C	12	B	22	B
3	A	13	B	23	B
4	A	14	D	24	C
5	B	15	B	25	A
6	B	16	A	26	B
7	C	17	B	27	C
8	C	18	C	28	B
9	C	19	A	29	B
10	B	20	B	30	D

🎯 Scoring

- **27–30:** Excellent — interview-ready on Day 1 fundamentals
- **24–26:** Very good
- **20–23:** Good, but revise weak areas
- **15–19:** Needs revision
- **<15:** Re-study today's concepts before moving ahead

Important: Don't just memorize the answer key. For Data Scientist interviews, you should be able to explain **why** the answer is correct.

ChatGPT can make mistakes. Check important info.